

1. Administrative & Study Identification

Full Study Title: A Randomized, Double-Blind, Placebo-Controlled Clinical Trial to Assess the Safety, Tolerability, and Pharmacokinetics of MK-1167 Administered to Patients With Alzheimer's Disease Receiving Stable Donepezil Treatment **Short Title/Acronym:** Efficacy and Safety of MK-1167 in Participants With Alzheimer's Disease Dementia Taking Stable Donepezil Treatment (MK-1167-007) **Protocol Number:** MK-1167-007-02 **NCT Number:** NCT06285240 **Posting ID:** 1089974905016072149 **Sponsor Name:** Merck **Investigational Product:** MK-1167 (an investigational oral positive allosteric modulator of the alpha-7 nicotinic acetylcholine receptor) **Phase of Development:** Phase 1 **Trial Status:** COMPLETED **Medical Monitor:** Merck Clinical Operations / Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate the safety and tolerability of MK-1167 when administered to patients with mild to moderate Alzheimer's Disease on stable doses of oral donepezil 10 mg daily. **Secondary Objectives:** To evaluate the pharmacokinetic (PK) profile of MK-1167 when administered as an adjunctive therapy alongside stable donepezil treatment.

2.2 Background & Rationale To address the need for innovative treatments targeting Alzheimer's Disease dementia. MK-1167 is designed to function as an adjunctive therapy to standard acetylcholinesterase inhibitors (AChEIs). By targeting and modulating the alpha-7 nicotinic acetylcholine receptors, MK-1167 potentially enhances the effectiveness of AChEIs like donepezil, addressing the loss of function in Alzheimer's patients and aiming to improve memory and cognitive skills.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized **Status:** Completed

3.2 Planned Enrollment & Duration **Targeted Enrollment:** 28 participants **Number of Sites:** Multicenter **Estimated Study Start:** Early 2024 **Estimated Primary Completion:** Late 2024 / 2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Reports a history of cognitive and functional decline with gradual onset and slow progression for at least 1 year before Screening, that is either corroborated by an informant who knows the subject well or is documented in medical records. 2. Meets the criteria for a diagnosis of probable Alzheimer's disease (AD) based on the National Institute of Neurological and Communicative Disorders - Stroke and the Alzheimer's Disease and Related Disorders Association (NINCDS-ADRDA) criteria for probable AD. 3. Is receiving donepezil 10 mg daily for symptomatic treatment of cognitive impairment associated with AD. The dose level must be stable for at least 2 months prior to Screening. If receiving donepezil via a transdermal system (ie, patch), it should be a 10-mg/day dose and should switch prescription to a 10-mg oral daily dose, before enrollment. 4. Has a reliable and competent trial partner/caregiver who has a close relationship with the participant, has face-to-face contact at least 3 days a week for a minimum of 6 waking hours a week, and is willing to accompany the participant, if desired, to study visits.

4.2 Exclusion Criteria 1. History of clinically significant endocrine, gastrointestinal, cardiovascular, hematological, hepatic, immunological, renal, respiratory, genitourinary, or major neurological (including stroke and chronic seizures) abnormalities or diseases that are not under medical control over the past 2 months. 2. Has evidence of a clinically relevant or unstable psychiatric disorder, based on DSM-5 criteria, or has a history of clinically significant psychiatric disorder in the last 5 years. Generalized anxiety disorder, and/or insomnia under good control for ≥ 2 months on stable medical therapy may not be exclusionary. 3. History of cancer (malignancy). Participants with adequately treated disease deemed as "cured," or who, in the opinion of the study investigator, are highly unlikely to sustain a recurrence for the duration of the study, may be enrolled at the discretion of the investigator and Sponsor. 4. History of significant multiple and/or severe allergies (eg, food, drug, latex allergy), or has had an anaphylactic reaction or significant intolerability (ie, systemic allergic reaction) to prescription or nonprescription drugs or food. 5. Had a major surgery and/or donated or lost 1 unit of blood (approximately 500 mL) within 4 weeks prior to the prestudy (screening) visit. 6. Unable to refrain from or anticipates the use of any medication, including prescription and nonprescription drugs or herbal remedies beginning approximately 2 weeks (or 5 half-lives) prior to administration of the initial dose of study intervention, throughout the study, until the poststudy visit. There may be certain protocol-specified medications that are permitted. 7. The participant is a smoker and/or has used nicotine or nicotine-containing products (eg, nicotine patch and electronic cigarette) within 3 months of screening. 8. Consumes greater than 3 servings of alcoholic beverages per day. Participants who consume 4

servings of alcoholic beverages per day may be enrolled at the discretion of the investigator. 9. The participant is a regular user of cannabis, any illicit drugs or has a history of drug (including alcohol) abuse within approximately 2 years.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Panel A:** Loading dose of MK-1167 6 mg (or matching placebo) once daily (QD) for Days 1 to 7, followed by MK-1167 3 mg (or matching placebo) QD for 14 consecutive days (Days 8 to 21). * **Panel B:** MK-1167 6 mg (or matching placebo) QD for Days 1 to 31. **Route of Administration:** Oral **Duration of Treatment:** 21 days for Panel A; 31 days for Panel B.

5.2 Concomitant & Prohibited Therapy Permitted: Stable dosing of donepezil 10 mg daily. **Prohibited:** Any medication (prescription, nonprescription, or herbal) beginning approximately 2 weeks prior to initial dose, unless explicitly permitted by the protocol.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Up to -28 Days	Informed Consent, Medical History, AD Diagnosis Confirmation
Baseline	Day 1	Randomization, First Dose, Extensive PK Sampling
Interim Visits	Days 8, 21 (Panel A) / Day 31 (Panel B)	Maintenance Dosing Adjustments, PK Blood Sampling, Safety & Tolerability Assessments
End of Study	Up to Week 7	Final Safety Checks, Adverse Event Tracking, PK Completion

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints

- 1. Number of Participants Who Experienced an Adverse Event (AE):** An AE was any untoward medical occurrence in a clinical study participant, temporally associated with the use of study intervention, whether or not considered related to the study intervention. An AE can therefore be any unfavorable and unintended sign (including an abnormal laboratory finding), symptom, or disease (new or exacerbated) temporally associated with the use of a study intervention. The number of participants who experienced an AE were reported.
- 2. Number of Participants Who Discontinued Study Treatment Due to an AE:** An AE was any untoward medical occurrence in a clinical study participant, temporally associated with the use of study intervention, whether or not considered related to the study intervention. An AE can therefore be any unfavorable and unintended sign (including an abnormal laboratory finding), symptom, or disease (new or exacerbated) temporally associated with the use of a study

intervention. The number of participants who discontinued study treatment due to an AE were reported.

7.2 Secondary Endpoints 1. **Panel A:** Area Under the Plasma Concentration-Time Curve From 0 to 24 Hours ($\$AUC_{\{0-24\}}\$$) After Administration of 6mg of MK-1167. 2. **Panel A:** $\$AUC_{\{0-24\}}\$$ After Administration of 3mg of MK-1167. 3. **Panel B:** $\$AUC_{\{0-24\}}\$$ After Administration of 6mg of MK-1167.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: AEs collected rigorously over the course of treatment and for an additional safety follow-up period (up to 7 weeks post-first dose). **Serious Adverse Events (SAEs):** Routinely monitored; any sign, symptom, or disease temporally associated with study intervention reported dynamically. **Data Monitoring Committee (DMC):** Panel B initiation strictly depends upon internal review of safety and tolerability data from Panel A participants.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all randomized participants receiving ≥ 1 dose of study drug); PK Parameter Set for serum concentration analyses. **Sample Size Justification:** A targeted enrollment of 28 participants deemed sufficient to establish early Phase 1 safety, tolerability, and PK parameter projections. **Handling of Missing Data:** Noncompartmental analysis for pharmacokinetic modeling.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** High-frequency onsite oversight particularly focused around intense PK sampling days (Day 1, 8, 21, 31). **Audit Trail:** Continuous review of electronic Case Report Forms (eCRF) alongside central laboratory pharmacokinetic evaluations.

11. Study Identifiers & Governance

Primary ID: $\{\{MK-1167-007-02\}\}$ **NCT Number:** $\{\{NCT06285240\}\}$
Posting ID: $\{\{1089974905016072149\}\}$ **Sponsor/Lead Organization:** $\{\{Merck\}\}$ **Study Title:** $\{\{Efficacy and Safety of MK-1167 in Participants With Alzheimer's Disease Dementia Taking Stable Donepezil Treatment (MK-1167-007)\}\}$ **Official Regulatory Title:** $\{\{A Randomized, Double-Blind, Placebo-Controlled Clinical Trial to Assess the Safety, Tolerability, and Pharmacokinetics of$

12. Clinical Framework (Alzheimer's Disease Specific)

Primary Condition: {{Alzheimer's Disease Dementia}} Disease
Areas: {{Alzheimer's Disease}} Diagnosis Threshold: {{Probable AD
based on NINCDS-ADRDA criteria}} Medical Methodology:
{{Acetylcholinesterase Inhibitor (AChEI) Base Therapy}}
Optimization Target: {{Adjuvant \$\\alpha\$7 Nicotinic Receptor
Modulation}}

13. Study Procedures & Methodology

Management Pathway: {{Adjunctive investigational therapy to
standard AD clinical pathway}} Education Protocol (HCP): {{Dose
titration and PK sampling schedule adherence}} Education Protocol
(Patient): {{Caregiver/Partner requirement for daily monitoring
assistance}} Quality Audit Mechanism: {{Intensive PK monitoring &
phased Panel B initiation based on safety data}}

14. Observation & Follow-Up Schedule

Total Duration: {{Up to approximately 7 weeks overall per
participant}} Onsite Visit Cadence: {{High frequency focusing on
Days 1, 8, 21, 23, and 31 depending on Panel}} Remote Monitoring
Frequency: {{Caregiver-assisted AE monitoring continuously}}
Checklist Review Interval: {{Weekly safety evaluation during the
active dosing period}}

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				